

Final Project

PSTAT122: Design and Analysis of Experiments

Fall 2025

STUDENT NAMES

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Due Date

Due Date: Monday, December 8, 2025, 11:59 PM

1 Introduction

Productivity during academic tasks is commonly influenced by various study behaviors, yet the extent to which these behaviors improve performance remains unclear. This study investigates how two conventional study treatments—listening to music and chewing gum—affect performance on a cognitive–motor task. To operationalize “productivity,” we narrowed the scope of our research question to examine measurable performance outcomes on a typing speed test.

Typing speed (measured in words per minute, WPM) was selected as the response variable because it represents a skill-based task that can be completed repeatedly within a short time frame, making it suitable for controlled experimentation. Although we initially planned to evaluate the effect of caffeine consumption on typing performance, practical constraints related to caffeine metabolism and required waiting periods made this approach infeasible for an in-session factorial experiment.

Therefore, our study focuses on the following research question: **Is there an interaction effect between listening to music and chewing gum on typing speed?** In addition to evaluating potential interaction effects, we also assess whether each factor independently influences performance.

This question is motivated by common student behaviors. Many students rely on music, fidgets, or small sensory stimuli (such as chewing gum) to improve concentration during extended study periods. By systematically testing these factors in a controlled setting, this experiment aims to determine whether these commonly used strategies enhance, hinder, or have no measurable effect on performance in a specific cognitive–motor task. We acknowledge that the effectiveness of sensory stimuli may vary across different types of tasks; however, this study seeks to provide preliminary evidence that may offer at least some broader insight into how such strategies influence academic performance.

2 Experimental Design

The response variable for this experiment was words per minute (WPM) recorded using Monkeytype. This platform was selected because its scoring system incorporates accuracy penalties, thereby providing a more reliable measure of typing performance than raw speed alone. The two study treatments—listening to music (Yes/No) and chewing gum (Yes/No)—were arranged in a full 2×2 factorial design, resulting in four treatment combinations. All participants used some sort of earpiece device to listen to music keeping this method consistent across participants to reduce variability.

Participants were allowed to choose the type of music they typically listen to in a productive setting. Although music genres varied across individuals, we treated “listening to music” as a single categorical factor to maintain feasibility and focus, as the experiment was not designed to examine genre-specific effects. The same principle was applied to chewing gum: participants used their preferred flavor, but flavor differences were not considered as part of the experimental factors.

All participants completed the typing task under standardized Monkeytype default settings to maintain consistency across trials. WPM values were recorded for each thirty second run, with the platform automatically adjusting scores downward when accuracy declined. To reduce bias, each participant’s order of treatment conditions was randomized. Within each assigned condition, participants completed ten consecutive typing trials, yielding ten WPM observations per condition. This replication served to reduce the influence of outliers and increase the precision of the estimated treatment effects.

Participants were treated as blocks in the design to account for inherent differences in baseline typing proficiency. With four participants and four conditions, this resulted in a total of 40 observations per condition and 160 observations overall. The experiment was completed in under one hour. To minimize fatigue effects, participants took 15-minute breaks between conditions. Thus, for each participant, the ten trials within a condition constitute ten replicates of the corresponding treatment combination.

3 Data Collection

Procedure & Challenges/Adjustments

The experiment was conducted remotely within a one-hour window due to participants being out of town and unable to meet in person. Each participant completed the study in a private, distraction-free environment to minimize uncontrolled external influences. For each assigned treatment condition, participants completed ten typing runs, recording their WPM score after each run. Upon completing a condition, participants were given a 15-minute break before moving on to the next randomized treatment.

One practical challenge of the study was the inability to gather all participants in the same physical location. As a result, clear instructions emphasized the importance of selecting a quiet, comfortable work space to ensure that environmental variability did not confound the effects of the experimental factors. In addition, participants chose their own keyboards of choice for comfortably. This is to emphasize and test participants writing speeds with already existing tools they are accustomed to.

Data Presentation

	Participant	Music	Gum	Run1	Run2	Run3	Run4	Run5	Run6	Run7	Run8	Run9	Run10	Avg
3	Isaac	No	Yes	62	51	54	55	63	61	60	64	54	60	59.4
2	Isaac	Yes	No	66	61	53	67	62	62	63	62	66	66	62.1

	Participant	Music	Gum	Run1	Run2	Run3	Run4	Run5	Run6	Run7	Run8	Run9	Run10	Avg
4	Isaac	Yes	Yes	60	56	46	54	58	55	64	73	59	66	68.0
1	Isaac	No	No	65	62	58	60	61	60	60	66	60	65	59.5
31	Tommy	No	Yes	57	51	54	84	79	90	68	81	82	87	91.3
11	Tommy	No	No	91	65	94	76	76	88	69	78	86	76	79.9
21	Tommy	Yes	No	94	86	81	81	79	74	71	78	86	70	78.0
41	Tommy	Yes	Yes	82	84	82	83	81	74	80	77	83	84	76.8
22	Chris	Yes	No	78	79	96	90	89	82	97	93	90	90	98.1
12	Chris	No	No	82	76	69	86	91	85	90	58	65	83	78.6
42	Chris	Yes	Yes	82	91	87	88	97	79	87	84	84	80	88.9
32	Chris	No	Yes	82	82	74	84	88	87	70	81	74	70	81.0
43	Christian	Yes	Yes	60	66	58	60	64	56	62	59	66	72	62.5
13	Christian	No	No	66	65	58	65	70	67	62	44	67	66	62.0
33	Christian	No	Yes	72	49	80	70	56	59	58	56	64	60	90.4
23	Christian	Yes	No	62	65	67	62	67	67	63	56	67	71	63.7

Table 2: Summary Statistics for Each Treatment Combination (Music $\times$ Gum)

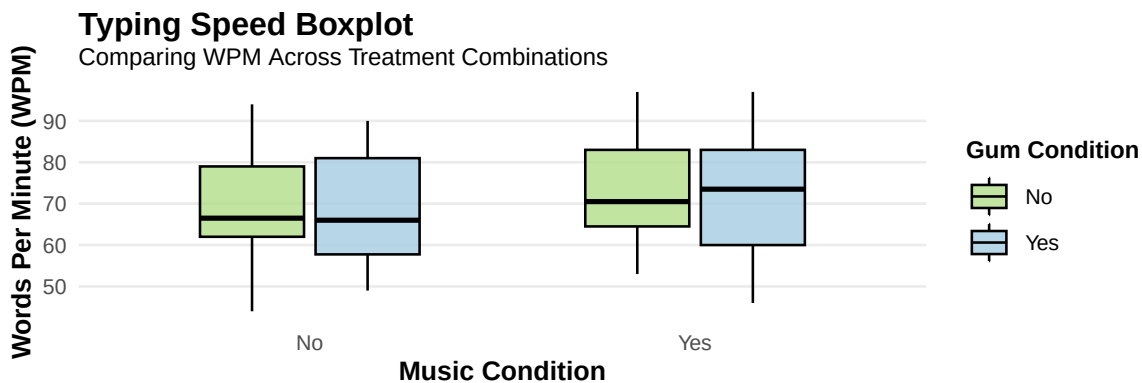
Music	Gum	Mean_WPM	SD_WPM	N_Runs	Min_WPM	Max_WPM
No	No	70.78	11.71	40	44	94
No	Yes	68.33	12.43	40	49	90
Yes	No	73.97	12.14	40	53	97
Yes	Yes	72.08	12.81	40	46	97

Table 3: Summary Statistics by Participant (Block) and Treatment

Participant	Music	Gum	Mean_WPM	SD_WPM	N_Runs	Min_WPM	Max_WPM
Chris	No	No	78.5	11.15	10	58	91
Chris	No	Yes	79.2	6.70	10	70	88
Chris	Yes	No	88.4	6.65	10	78	97
Chris	Yes	Yes	85.9	5.38	10	79	97
Christian	No	No	63.0	7.41	10	44	70
Christian	No	Yes	62.4	9.19	10	49	80
Christian	Yes	No	64.7	4.14	10	56	71
Christian	Yes	Yes	62.3	4.76	10	56	72
Isaac	No	No	61.7	2.71	10	58	66
Isaac	No	Yes	58.4	4.50	10	51	64
Isaac	Yes	No	62.8	4.08	10	53	67
Isaac	Yes	Yes	59.1	7.39	10	46	73
Tommy	No	No	79.9	9.52	10	65	94
Tommy	No	Yes	73.3	14.58	10	51	90
Tommy	Yes	No	80.0	7.39	10	70	94
Tommy	Yes	Yes	81.0	3.23	10	74	84

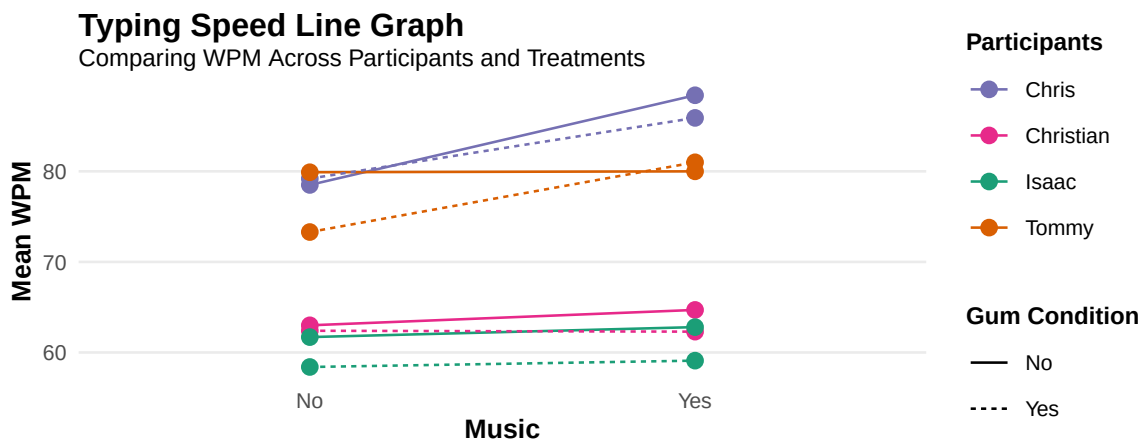
4 Analysis

Exploratory Data



Initially, the data leads to the initial observation that neither the gum nor music treatments appear to have a strong effect on the average WPM. This is primarily evident in the above box plot, which displays the four means (corresponding to the four treatment conditions) as being all within each others' standard deviations. However, there is a slight indication that music may have a positive effect on WPM, as the two "Yes" music conditions have marginally higher means than their "No" music counterparts.

All four of the means are relatively close together, with the highest mean (73.97 WPM for Yes Music and No Gum) being only about 5 WPM higher than the lowest mean (68.33 WPM for No Music and Yes Gum). Additionally, all of the observations have relatively high standard deviations (all of which hover around 12 WPM), indicating a high level of variability in typing speeds across all treatment conditions. This suggests that individual differences between participants may be a more significant factor in typing speed than the treatments themselves.



Upon looking at the line graph above, we can see that there are substantial individual differences in typing speed among the four participants. Chris and Tommy consistently type at much higher speeds than Christian and Isaac across all treatment conditions. This suggests that individual baseline typing ability may play a more significant role in determining typing speed than the experimental treatments of music and gum. These strong block effects indicate that **participant variability is far larger than any music or gum effect**, and any conclusions about treatment differences must account for these baseline performance differences. Despite this, no definitive conclusions can be drawn from these initial observations alone, and further statistical analysis is necessary to determine the significance of any observed effects.

Hypothesis Testing

Before we conduct the ANOVA, we must first ensure that the assumptions of the test are met. The independence assumption is satisfied as each run was randomized for each participant, and there were breaks between runs to minimize fatigue or practice effects. The normality assumption is reasonably met, as indicated by the QQ plot and the Shapiro-Wilk test, which shows that while the data is not perfectly normal, the ANOVA is robust to moderate violations of normality with a sufficiently large sample size ($n=160$). The homoscedasticity assumption is also satisfied, as shown by the residuals vs. fitted plot and the Bartlett test, which indicates homogeneity of variances across treatment groups(all of the R processes used for validating the aforementioned assumptions can be found in the *Tables, Figures, & Code* section). We then conducted AIC and BIC model selection procedures to identify the most appropriate model for our data(these processes can be found in the *Tables, Figures, & Code* section). Both methods suggested that the interaction term between music and gum did not significantly improve the model fit, leading us to exclude it from the final model. **The final model followed: $WPM \sim \text{music} + \text{gum} + \text{participants}$.** We establish the following hypotheses:

$$\begin{aligned} H_0 : \beta_{\text{music}} &= 0 \\ H_1 : \beta_{\text{music}} &\neq 0 \\ H_0 : \beta_{\text{gum}} &= 0 \\ H_1 : \beta_{\text{gum}} &\neq 0 \end{aligned} \tag{1}$$

We test these hypotheses using the following ANOVA table:

Table 4: Regression Results for $WPM \sim \text{music} + \text{gum} + \text{participants}$

term	estimate	std.error	statistic	p.value	conf.low	conf.high
(Intercept)	82.350	1.446	56.942	0.000	79.493	85.207
musicYes	3.475	1.181	2.943	0.004	1.142	5.808
gumYes	-2.175	1.181	-1.842	0.067	-4.508	0.158
participantsChristian	-19.900	1.670	-11.917	0.000	-23.199	-16.601
participantsIsaac	-22.500	1.670	-13.473	0.000	-25.799	-19.201
participantsTommy	-4.450	1.670	-2.665	0.009	-7.749	-1.151

Based on the aforementioned ANOVA results, we find that the p-value for the music treatment is 0.004, which is less than our significance level of 0.05. Therefore, we reject the null hypothesis for music and conclude that listening to music has a statistically significant effect on typing speed. On the other hand, the p-value for the gum treatment is 0.067, which is greater than 0.05. Thus, we fail to reject the null hypothesis for gum and conclude that chewing gum does not have a statistically significant effect on typing speed. Additionally, the three participants also show that they have statistically significant typing speeds than the baseline participant(Chris). This indicates that individual differences between participants play a significant role in typing speed, further emphasizing the importance of accounting for participant variability in our analysis.

Essentially, we are able to conclude the following:

- Listening to music has a statistically significant positive effect on typing speed. In our experiment, participants who listened to music typed, on average, about 3.5 WPM faster than those who did not listen to music.
- Chewing gum does not have a statistically significant effect on typing speed. The data suggests a slight decrease in WPM when chewing gum, but this effect is only significant at the 0.1 level, not the required 0.05 level. Therefore, this conclusion is marginal, but we cannot confidently state that gum has a meaningful impact on typing speed.
- There is no significant interaction effect between music and gum on typing speed. The lack of a significant interaction term in our model indicates that the effects of music and gum on typing speed are independent

of each other.

- The largest source of variation and significance in typing speed comes from individual differences between participants. The participants themselves were consistent around their own average WPM, but there were large differences in average WPM between participants. This suggests that inherent typing ability or other individual factors play a more substantial role in determining typing speed than the experimental treatments.

Tables, Figures, & Code:

Validity of the ANOVA Assumptions

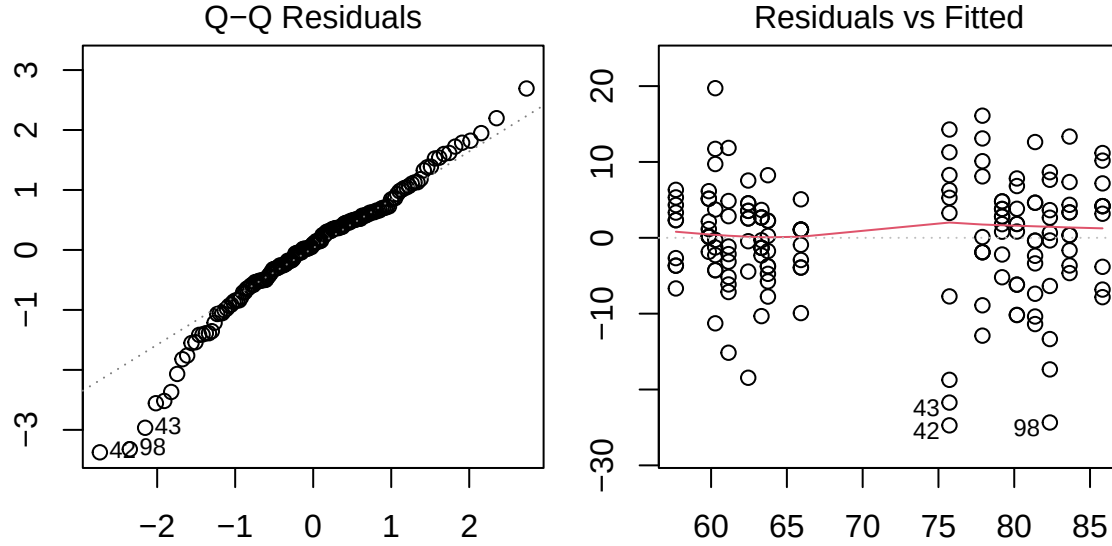


Figure 1: ANOVA Diagnostic Plots (Q-Q and Residuals vs Fitted)

Table 5: Shapiro–Wilk Normality Test

	Statistic	P_Value
W	0.96554	0.0005095

Table 6: Bartlett Test for Homogeneity of Variances

	K_Squared	df	P_Value
Bartlett's K-squared	0.32859	3	0.95456

AIC and BIC Model Selection

Table 7: Stepwise Model Selection for WPM

Step	Model	AIC	RSS
Start	WPM ~ music + gum + music*gum + participants	651.24	8586.2
Remove music	WPM ~ music + gum + participants	649.29	8589.2
Remove gum	WPM ~ music + participants	650.78	8778.5

Step	Model	AIC	RSS
Remove music	WPM ~ gum + participants	656.05	9072.3
Remove participants	WPM ~ music + gum	804.49	23522.5

Additional Code Found in Appendices

The following code is organized and found in the appendices section in the following order:

- 6.1) Summary Statistics from Section 3(Data Collection~Data Presentation)
- 6.2) Box Plot and Line Graph from Section 4(Analysis~Exploratory Data)
- 6.3) Regression Table from Section 4(Analysis~Hypothesis Testing)
- 6.4) Assumption Graphs from Section 4(Analysis~Tables, Figures, & Code)
- 6.5) Assumption Tables from Section 4(Analysis~Tables, Figures, & Code)
- 6.6) AIC and BIC Model Selection from Section 4(Analysis~Tables, Figures, & Code)
- 6.7) Initial Data Frame of Experiment Results
- 6.8) ANOVA Model Using Averages(instead of raw data)
- 6.9) ANOVA Model Using Raw Data(before AIC/BIC optimization)
- 6.10) Full AIC/BIC Model Selection
- 6.11) Interaction Plot Visual

[6.1 - 6.6 are snippets of code that were used in the report's sections 1-5; the output of these snippets are hidden in the appendix, but visible in the report sections]

[6.7 - 6.11 are snippets of code that were NOT used in the report's sections 1-5; the output of these snippets are visible in the appendix]

5 Conclusions

- Based on the ANOVA results, we conclude that listening to music has a statistically significant positive effect on typing speed, while chewing gum does not have a statistically significant effect. There is no significant interaction effect between music and gum on typing speed. The largest source of variation in typing speed comes from individual differences between participants.
- However, due to the nature of our experimentation, these results can only be generalized to similar individuals performing similar tasks to the task in our experiment. Further research with a larger and more diverse sample size would be necessary to generalize these findings to a broader population. However, it is safe to say that for the participants in our study, listening to music had a significant effect by increasing the typing speed by ~3.5 WPM.
- The results of our experiments can be attributed to a number of possible reasons. For example, music may have had a positive, significant effect on typing speed since the participants could choose music they would typically study to, which may have enhanced their focus on the task at hand. Adversely, the music could have provided rhythm that the participants could have subconsciously followed while typing, leading to increased speed. On the other hand, chewing gum may not have had a significant effect because it could have been a distraction for some participants, or the act of chewing may not have provided any cognitive benefits that would translate to improved typing performance.
- Our experiment was limited in its findings mostly to the small sample size and the inability to control certain confounding variables. Despite our randomization, blocking, and replication of trials, there is no way to control the fact that people all have different inherent typing speeds and styles. Additionally, if this experiment were to be repeated, certain confounding variables should be controlled for, such as the type of music the participants listened to, the manufacturer of the keyboards used,

the gum flavor and brand, the time of day the experiment was conducted, and the environment in which the participants completed the task. Future studies could also explore different types of tasks beyond typing speed to see if the effects of music and gum vary across different cognitive or motor activities.

6 Appendices

6.1) Summary Statistics from Section 3(Data Collection~Data Presentation)

```
1 library(tidyverse)
2 library(knitr)
3 set.seed(123)
4
5 participants <- c("Chris", "Christian", "Isaac", "Tommy")
6 conditions <- data.frame(
7   Music = c("No", "Yes", "No", "Yes"),
8   Gum    = c("No", "No", "Yes", "Yes")
9 )
10
11
12 randomize_conditions <- function(name) {
13   cond <- conditions[sample(1:4), ]
14   cond$Participant <- name
15   cond
16 }
17
18 design_list <- lapply(sample(participants), randomize_conditions)
19 design_df <- do.call(rbind, design_list)
20
21 design_df$Run1 <- c(62, 66, 60, 65, 57, 91, 94, 82, 78, 82, 82, 82, 60, 66, 72, 62)
22 design_df$Run2 <- c(51, 61, 56, 62, 51, 65, 86, 84, 79, 76, 91, 82, 66, 65, 49, 65)
23 design_df$Run3 <- c(54, 53, 46, 58, 54, 94, 81, 82, 96, 69, 87, 74, 58, 58, 80, 67)
24 design_df$Run4 <- c(55, 67, 54, 60, 84, 76, 81, 83, 90, 86, 88, 84, 60, 65, 70, 62)
25 design_df$Run5 <- c(63, 62, 58, 61, 79, 76, 79, 81, 89, 91, 97, 88, 64, 70, 56, 67)
26 design_df$Run6 <- c(61, 62, 55, 60, 90, 88, 74, 74, 82, 85, 79, 87, 56, 67, 59, 67)
27 design_df$Run7 <- c(60, 63, 64, 60, 68, 69, 71, 80, 97, 90, 87, 70, 62, 62, 58, 63)
28 design_df$Run8 <- c(64, 62, 73, 66, 81, 78, 78, 77, 93, 58, 84, 81, 59, 44, 56, 56)
29 design_df$Run9 <- c(54, 66, 59, 60, 82, 86, 86, 83, 90, 65, 84, 74, 66, 67, 64, 67)
30 design_df$Run10 <- c(60, 66, 66, 65, 87, 76, 70, 84, 90, 83, 80, 70, 72, 66, 60, 71)
31 design_df$Avg <- c(59.4, 62.1, 68, 59.5, 91.3, 79.9, 78, 76.8, 98.1, 78.6, 88.9, 81,
32   62.5, 62, 90.4, 63.7)
33
34 design_df <- design_df[, c("Participant", "Music", "Gum",
35   "Run1", "Run2", "Run3", "Run4", "Run5", "Run6", "Run7",
36   "Run8", "Run9", "Run10", "Avg")]
37
```

```

38 knitr::kable(design_df)
39
40
41
42 # Convert wide → long format for summary statistics
43 long_df <- design_df |>
44   pivot_longer(
45     cols = starts_with("Run"),
46     names_to = "Run",
47     values_to = "WPM"
48   )
49
50 # Compute summary statistics by Music × Gum
51 treatment_summary <- long_df |>
52   group_by(Music, Gum) |>
53   summarise(
54     Mean_WPM = mean(WPM),
55     SD_WPM   = sd(WPM),
56     N_Runs   = n(),
57     Min_WPM  = min(WPM),
58     Max_WPM  = max(WPM),
59     .groups  = "drop"
60   )
61
62 # Knit summary table
63 kable(
64   treatment_summary,
65   caption = "Summary Statistics for Each Treatment Combination (Music × Gum)",
66   digits = 2,
67   align = "c"
68 )
69
70 # Summary by participant block (Participant × Music × Gum)
71 participant_summary <- long_df |>
72   group_by(Participant, Music, Gum) |>
73   summarise(
74     Mean_WPM = mean(WPM),
75     SD_WPM   = sd(WPM),
76     N_Runs   = n(),
77     Min_WPM  = min(WPM),
78     Max_WPM  = max(WPM),
79     .groups  = "drop"
80   ) |>
81   arrange(Participant, Music, Gum)
82
83 kable(
84   participant_summary,
85   caption = "Summary Statistics by Participant (Block) and Treatment",
86   digits = 2,
87   align = "c"

```

88)

6.2) Box Plot and Line Graph from Section 4(Analysis~Exploratory Data)

```
1 library(ggplot2)
2 library(tidyr)
3 library(dplyr)
4
5 monkey_data_frame <- data.frame(
6   participants = rep(c("Isaac", "Tommy", "Chris", "Christian"), each = 4) ,
7
8   music = c("No", "Yes", "Yes", "No",
9             "No", "No", "Yes", "Yes",
10            "Yes", "No", "Yes", "No",
11            "Yes", "No", "No", "Yes") ,
12
13   gum = c("Yes", "No", "Yes", "No",
14           "Yes", "No", "No", "Yes",
15           "No", "No", "Yes", "Yes",
16           "Yes", "No", "Yes", "No") ,
17
18   run1 = c(62, 66, 60, 65, 57, 91, 94, 82, 78, 82, 82, 82, 60, 66, 72, 62),
19   run2 = c(51, 61, 56, 62, 51, 65, 86, 84, 79, 76, 91, 82, 66, 65, 49, 65),
20   run3 = c(54, 53, 46, 58, 54, 94, 81, 82, 96, 69, 87, 74, 58, 58, 80, 67),
21   run4 = c(55, 67, 54, 60, 84, 76, 81, 83, 90, 86, 88, 84, 60, 65, 70, 62),
22   run5 = c(63, 62, 58, 61, 79, 76, 79, 81, 89, 91, 97, 88, 64, 70, 56, 67),
23   run6 = c(61, 62, 55, 60, 90, 88, 74, 74, 82, 85, 79, 87, 56, 67, 59, 67),
24   run7 = c(60, 63, 64, 60, 68, 69, 71, 80, 97, 90, 87, 70, 62, 62, 58, 63),
25   run8 = c(64, 62, 73, 66, 81, 78, 78, 77, 93, 58, 84, 81, 59, 44, 56, 56),
26   run9 = c(54, 66, 59, 60, 82, 86, 86, 83, 90, 65, 84, 74, 66, 67, 64, 67),
27   run10 = c(60, 66, 66, 65, 87, 76, 70, 84, 90, 83, 80, 70, 72, 66, 60, 71),
28   average = c(59.4, 62.1, 68, 59.6, 91.3, 79.9, 78, 76.8, 98.1, 78.6, 88.9,
29              81, 62.5, 62, 90.4, 63.7)
30 )
31
32 long_data <- monkey_data_frame |>
33   pivot_longer(
34     cols = starts_with("run"),
35     names_to = "run",
36     values_to = "WPM"
37   )
38 tight_theme <- theme_minimal(base_size = 10, base_family = "sans") +
39   theme(
40     plot.title = element_text(face = "bold", size = 12, margin = margin(b = 2)),
41     plot.subtitle = element_text(size = 9, margin = margin(b = 4)),
42     axis.title = element_text(face = "bold", size = 10),
43     axis.text = element_text(size = 8),
44     legend.title = element_text(face = "bold", size = 9),
45     legend.text = element_text(size = 8),
46     panel.grid.minor = element_blank(),
```

```

47     panel.grid.major.x = element_blank(),
48     plot.margin = margin(t = 2, r = 2, b = 2, l = 2)
49   )
50
51   theme_set(tight_theme)
52
53   # participants color palette (used in line graph)
54   my_colors <- c(
55     "Isaac" = "#1b9e77",
56     "Tommy" = "#d95f02",
57     "Chris" = "#7570b3",
58     "Christian" = "#e7298a"
59   )
60
61   # gum fill palette (used in both)
62   gum_colors <- c(
63     "Yes" = "#a6cee3",
64     "No" = "#b2df8a"
65   )
66
67   monkey_boxplot <- ggplot(long_data, aes(x = music, y = WPM, fill = gum)) +
68     geom_boxplot(alpha = 0.8, width = 0.6, color = "black") +
69     scale_fill_manual(values = gum_colors) +
70     labs(
71       title = "Typing Speed Boxplot",
72       subtitle = "Comparing WPM Across Treatment Combinations",
73       x = "Music Condition",
74       y = "Words Per Minute (WPM)",
75       fill = "Gum Condition"
76     )
77
78   monkey_boxplot
79
80   participant_means <- long_data |>
81     group_by(participants, music, gum) |>
82     summarise(Mean_WPM = mean(WPM), .groups = "drop")
83
84   monkey_lineplot <- ggplot(
85     participant_means,
86     aes(
87       x = music,
88       y = Mean_WPM,
89       color = participants,
90       group = interaction(participants, gum),
91       linetype = gum
92     )
93   ) +
94     geom_line() +
95     geom_point(size = 2.5) +
96     scale_color_manual(values = my_colors) +

```

```

97   labs(
98     title = "Typing Speed Line Graph",
99     subtitle = "Comparing WPM Across Participants and Treatments",
100    x = "Music",
101    y = "Mean WPM",
102    color = "Participants",
103    linetype = "Gum Condition"
104  )
105
106 monkey_lineplot

```

6.3) Regression Table from Section 4(Analysis~Hypothesis Testing)

```

1  library(broom)
2  library(knitr)
3
4  model <- lm(WPM ~ music + gum + participants, data = long_data)
5
6  kable(
7    tidy(model, conf.int = TRUE),
8    digits = 3,
9    caption = "Regression Results for WPM ~ music + gum + participants",
10   align = "c"
11 )

```

6.4) Assumption Graphs from Section 4(Analysis~Tables, Figures, & Code)

```

1  model <- lm(WPM ~ music + gum + participants, data = long_data)
2  par(mfrow = c(1,2), mar = c(2,2,2,1), oma = c(0,0,0,0), asp=NA)
3  plot(model, which = 2)
4  plot(model, which = 1)
5  par(mfrow = c(1,1))

```

6.5) Assumption Tables from Section 4(Analysis~Tables, Figures, & Code)

```

1  sw <- shapiro.test(residuals(model))
2
3  sw_table <- data.frame(
4    Statistic = round(sw$statistic, 5),
5    P_Value   = signif(sw$p.value, 5)
6  )
7
8  knitr::kable(sw_table, align = "c", caption = "Shapiro-Wilk Normality Test")
9  long_data$group <- paste(long_data$music, long_data$gum, sep = "_")
10
11 bt <- bartlett.test(WPM ~ group, data = long_data)
12
13 bt_table <- data.frame(

```

```

14   K_Squared = round(bt$statistic, 5),
15   df        = bt$parameter,
16   P_Value   = signif(bt$p.value, 5)
17 )
18
19 knitr::kable(bt_table,
20               align = "c",
21               caption = "Bartlett Test for Homogeneity of Variances")

```

6.6) AIC and BIC Model Selection from Section 4(Analysis~Tables, Figures, & Code)

```

1  library(knitr)
2  model_summary <- data.frame(
3    Step = c("Start", "Remove music", "Remove gum", "Remove music", "Remove participants"),
4    Model = c(
5      "WPM ~ music + gum + music*gum + participants",
6      "WPM ~ music + gum + participants",
7      "WPM ~ music + participants",
8      "WPM ~ gum + participants",
9      "WPM ~ music + gum"
10   ),
11   AIC = c(651.24, 649.29, 650.78, 656.05, 804.49),
12   RSS = c(8586.2, 8589.2, 8778.5, 9072.3, 23522.5)
13 )
14
15 kable(model_summary, digits = 2, caption = "Stepwise Model Selection for WPM")

```

6.7) Initial Data Frame of Experiment Results

```

1  library(dbplyr)
2  library(tidyr)
3  library(tidyverse)
4  library(DescTools)
5
6  # 1) DATAFRAME:
7  monkey_data_frame <- data.frame(
8    participants = rep(c("Isaac", "Tommy", "Chris", "Christian"), each = 4) ,
9
10   music = c("No", "Yes", "Yes", "No",
11             "No", "No", "Yes", "Yes",
12             "Yes", "No", "Yes", "No",
13             "Yes", "No", "No", "Yes") ,
14
15   gum = c("Yes", "No", "Yes", "No",
16           "Yes", "No", "No", "Yes",
17           "No", "No", "Yes", "Yes",
18           "Yes", "No", "Yes", "No") ,
19
20   run1 = c(62, 66, 60, 65, 57, 91, 94, 82, 78, 82, 82, 82, 60, 66, 72, 62),

```

```

21     run2 = c(51, 61, 56, 62, 51, 65, 86, 84, 79, 76, 91, 82, 66, 65, 49, 65),
22     run3 = c(54, 53, 46, 58, 54, 94, 81, 82, 96, 69, 87, 74, 58, 58, 80, 67),
23     run4 = c(55, 67, 54, 60, 84, 76, 81, 83, 90, 86, 88, 84, 60, 65, 70, 62),
24     run5 = c(63, 62, 58, 61, 79, 76, 79, 81, 89, 91, 97, 88, 64, 70, 56, 67),
25     run6 = c(61, 62, 55, 60, 90, 88, 74, 74, 82, 85, 79, 87, 56, 67, 59, 67),
26     run7 = c(60, 63, 64, 60, 68, 69, 71, 80, 97, 90, 87, 70, 62, 62, 58, 63),
27     run8 = c(64, 62, 73, 66, 81, 78, 78, 77, 93, 58, 84, 81, 59, 44, 56, 56),
28     run9 = c(54, 66, 59, 60, 82, 86, 86, 83, 90, 65, 84, 74, 66, 67, 64, 67),
29     run10 = c(60, 66, 66, 65, 87, 76, 70, 84, 90, 83, 80, 70, 72, 66, 60, 71),
30     average = c(59.4, 62.1, 68, 59.6, 91.3, 79.9, 78, 76.8, 98.1, 78.6, 88.9,
31                 81, 62.5, 62, 90.4, 63.7)
32 )
33
34 monkey_data_frame

```

	participants	music	gum	run1	run2	run3	run4	run5	run6	run7	run8	run9	run10
1	Isaac	No	Yes	62	51	54	55	63	61	60	64	54	60
2	Isaac	Yes	No	66	61	53	67	62	62	63	62	66	66
3	Isaac	Yes	Yes	60	56	46	54	58	55	64	73	59	66
4	Isaac	No	No	65	62	58	60	61	60	60	66	60	65
5	Tommy	No	Yes	57	51	54	84	79	90	68	81	82	87
6	Tommy	No	No	91	65	94	76	76	88	69	78	86	76
7	Tommy	Yes	No	94	86	81	81	79	74	71	78	86	70
8	Tommy	Yes	Yes	82	84	82	83	81	74	80	77	83	84
9	Chris	Yes	No	78	79	96	90	89	82	97	93	90	90
10	Chris	No	No	82	76	69	86	91	85	90	58	65	83
11	Chris	Yes	Yes	82	91	87	88	97	79	87	84	84	80
12	Chris	No	Yes	82	82	74	84	88	87	70	81	74	70
13	Christian	Yes	Yes	60	66	58	60	64	56	62	59	66	72
14	Christian	No	No	66	65	58	65	70	67	62	44	67	66
15	Christian	No	Yes	72	49	80	70	56	59	58	56	64	60
16	Christian	Yes	No	62	65	67	62	67	67	63	56	67	71
average													
1	59.4												
2	62.1												
3	68.0												
4	59.6												
5	91.3												
6	79.9												
7	78.0												
8	76.8												
9	98.1												
10	78.6												
11	88.9												
12	81.0												
13	62.5												
14	62.0												
15	90.4												
16	63.7												

6.8) ANOVA Model Using Averages(instead of raw data)

```
1 model_average <- lm(average ~ gum + music + gum*music + participants,  
2                       data = monkey_data_frame)  
3 average_anova_result <- anova(model_average)  
4 average_anova_result
```

Analysis of Variance Table

Response: average

	Df	Sum Sq	Mean Sq	F value	Pr(>F)
gum	1	82.36	82.36	0.9701	0.35039
music	1	1.05	1.05	0.0124	0.91386
participants	3	1474.08	491.36	5.7878	0.01742 *
gum:music	1	142.21	142.21	1.6751	0.22780
Residuals	9	764.06	84.90		

Signif. codes: 0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1

```
1 average_summary_result <- summary(model_average)  
2 print(average_summary_result)
```

Call:

```
lm(formula = average ~ gum + music + gum * music + participants,  
    data = monkey_data_frame)
```

Residuals:

Min	1Q	Median	3Q	Max
-11.156	-4.513	-1.644	3.619	15.244

Coefficients:

	Estimate	Std. Error	t value	Pr(> t)
(Intercept)	81.656	6.094	13.399	3e-07 ***
gumYes	10.500	6.515	1.612	0.14150
musicYes	5.450	6.515	0.837	0.42452
participantsChristian	-17.000	6.515	-2.609	0.02830 *
participantsIsaac	-24.375	6.515	-3.741	0.00462 **
participantsTommy	-5.150	6.515	-0.790	0.44958
gumYes:musicYes	-11.925	9.214	-1.294	0.22780

Signif. codes: 0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1

Residual standard error: 9.214 on 9 degrees of freedom

Multiple R-squared: 0.6899, Adjusted R-squared: 0.4831

F-statistic: 3.337 on 6 and 9 DF, p-value: 0.05148

6.9) ANOVA Model Using Raw Data(before AIC/BIC optimization)

```

1 long_data <- monkey_data_frame |>
2   pivot_longer(
3     cols = starts_with("run"),
4     names_to = "run",
5     values_to = "WPM" )
6 head(long_data, 50)

```

```

# A tibble: 50 x 6
  participants music gum   average run   WPM
  <chr>         <chr> <chr>   <dbl> <chr> <dbl>
1 Isaac       No    Yes    59.4 run1    62
2 Isaac       No    Yes    59.4 run2    51
3 Isaac       No    Yes    59.4 run3    54
4 Isaac       No    Yes    59.4 run4    55
5 Isaac       No    Yes    59.4 run5    63
6 Isaac       No    Yes    59.4 run6    61
7 Isaac       No    Yes    59.4 run7    60
8 Isaac       No    Yes    59.4 run8    64
9 Isaac       No    Yes    59.4 run9    54
10 Isaac      No    Yes    59.4 run10   60
# i 40 more rows

```

```

1 model_all <- lm(WPM ~ music + gum + music*gum + participants, data = long_data)
2 anova(model_all)

```

Analysis of Variance Table

Response: WPM

	Df	Sum Sq	Mean Sq	F value	Pr(>F)
music	1	483.0	483.0	8.6071	0.003864 **
gum	1	189.2	189.2	3.3718	0.068260 .
participants	3	14933.3	4977.8	88.6999	< 2.2e-16 ***
music:gum	1	3.0	3.0	0.0539	0.816716
Residuals	153	8586.2	56.1		

Signif. codes: 0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1

```

1 print(summary(model_all))

```

Call:

```

lm(formula = WPM ~ music + gum + music * gum + participants,
    data = long_data)

```

Residuals:

Min	1Q	Median	3Q	Max
-24.5875	-3.7875	0.5875	4.3125	19.8625

Coefficients:

	Estimate	Std. Error	t value	Pr(> t)
(Intercept)	82.487	1.567	52.643	< 2e-16 ***
musicYes	3.200	1.675	1.910	0.05796 .
gumYes	-2.450	1.675	-1.463	0.14563
participantsChristian	-19.900	1.675	-11.880	< 2e-16 ***
participantsIsaac	-22.500	1.675	-13.432	< 2e-16 ***
participantsTommy	-4.450	1.675	-2.657	0.00873 **
musicYes:gumYes	0.550	2.369	0.232	0.81672

Signif. codes: 0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1

Residual standard error: 7.491 on 153 degrees of freedom

Multiple R-squared: 0.6451, Adjusted R-squared: 0.6312

F-statistic: 46.36 on 6 and 153 DF, p-value: < 2.2e-16

6.10) Full AIC/BIC Model Selection

```
1 library(car)
2 model_all <- lm(WPM ~ music + gum + music*gum + participants, data = long_data)
3 step(model_all, direction = "both")
```

Start: AIC=651.24

WPM ~ music + gum + music * gum + participants

	Df	Sum of Sq	RSS	AIC
- music:gum	1	3	8589.2	649.29
<none>			8586.2	651.24
- participants	3	14933	23519.5	806.47

Step: AIC=649.29

WPM ~ music + gum + participants

	Df	Sum of Sq	RSS	AIC
<none>			8589.2	649.29
- gum	1	189.2	8778.5	650.78
+ music:gum	1	3.0	8586.2	651.24
- music	1	483.0	9072.3	656.05
- participants	3	14933.3	23522.5	804.49

Call:

lm(formula = WPM ~ music + gum + participants, data = long_data)

Coefficients:

(Intercept)	musicYes	gumYes
82.350	3.475	-2.175
participantsChristian	participantsIsaac	participantsTommy
-19.900	-22.500	-4.450

```

1 n <- nrow(long_data)
2 step(model_all, direction = "both", k =log(n))

```

Start: AIC=672.76

WPM ~ music + gum + music * gum + participants

	Df	Sum of Sq	RSS	AIC
- music:gum	1	3	8589.2	667.75
<none>			8586.2	672.76
- participants	3	14933	23519.5	818.77

Step: AIC=667.75

WPM ~ music + gum + participants

	Df	Sum of Sq	RSS	AIC
- gum	1	189.2	8778.5	666.16
<none>			8589.2	667.75
- music	1	483.0	9072.3	671.42
+ music:gum	1	3.0	8586.2	672.76
- participants	3	14933.3	23522.5	813.71

Step: AIC=666.16

WPM ~ music + participants

	Df	Sum of Sq	RSS	AIC
<none>			8778.5	666.16
+ gum	1	189.2	8589.2	667.75
- music	1	483.0	9261.5	669.65
- participants	3	14933.3	23711.7	809.92

Call:

lm(formula = WPM ~ music + participants, data = long_data)

Coefficients:

(Intercept)	musicYes	participantsChristian
81.262	3.475	-19.900
participantsIsaac	participantsTommy	
-22.500	-4.450	

6.11) Interaction Plot Visual

```

1 library(ggplot2)
2
3 mean_data <- long_data |>
4   group_by(music, gum) |>
5   summarise(Mean_WPM = mean(WPM))
6

```

```

7 interaction_plot <- ggplot(mean_data, aes(x = music, y = Mean_WPM,
8                                           color = gum, group = gum)) +
9   geom_line(size = 1.5) +
10  geom_point(size = 3) +
11  labs(
12    title = "Interaction Plot: Music × Gum Effects on Average Typing Speed",
13    x = "Music",
14    y = "Mean WPM" )
15
16 interaction_plot

```

